

Roll No.

TME-812

B. TECH. (ME) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2022
NON-CONVENTIONAL ENERGY RESOURCES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What do you mean by energy exploitation ? Explain in detail. (CO1)
- (b) What are different non-conventional energy resources ? Explain in detail. (CO1)
- (c) Explain energy demand analysis in detail. (CO1)
2. (a) What do you mean by solar radiation ? Explain the working principle of flat plate collector with neat sketch. (CO2)

P. T. O.

- (b) Write energy balance equations for different components of compound parabolic concentrator (CPC) and hence derive the expression for temperature at the outlet of CPC. (CO2)
- (c) Explain different sensible heat storage methods with neat sketch. (CO2)
3. (a) What are aerodynamic considerations in Windmill design ? Explain. (CO3)
- (b) A propelling type of wind turbine has the following data : (CO3)
- Speed of wind at the height of 10 m = 12 m/s.
- Air density = 1.226 kg/m^3
- $\alpha = 0.14$
- Height of tower = 100 m
- Diameter of rotor = 80 m,
- Wind speed at the turbine reduces by 20% and
- Generator efficiency = 85%.
- Find :
- (i) total power available in wing
 - (ii) power extracted by the turbine
 - (iii) electrical power generated
 - (iv) axial thrust on the turbine and
 - (v) maximum axial thrust on the turbine.
- Assume suitable data if required.

(3)

- (c) Explain the different types of geothermal station with schematic representation. (CO3)
4. (a) What is tidal power ? Explain the different components of tidal power plant. Write the limitations of tidal energy conversion systems. (CO4, CO5)
- (b) A tidal power plant of single basin type has basin area of 24 km^2 . The tide has a range of 10 m. The turbine stops operation when the head on it falls below 3 m. Calculate the average power generated during one filling/emptying in MW if the turbine generator efficiency is 75%. Density of sea water = 1025 kg/m^3 . Assume suitable data if required. (CO4, CO5)
- (c) Explain the different methods of production of hydrogen with neat sketch in short. (CO4, CO5)
5. (a) What is energy audit ? Explain the energy audit for boilers and turbines. (CO6)
- (b) Write in detail the various applications of hydrogen energy. (CO6)
- (c) Explain the different types of horizontal axis wind turbine power plant. (CO6)

Roll No.

TME-814

B. TECH. (ME) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2022
ADVANCED WELDING TECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define welding. Classify welding processes based on source of energy used. (CO1)
- (b) What are different joining processes ? Compare welding joint with rivetted joint. (CO1)
- (c) Draw schematic diagram of oxyacetylene gas welding process and explain working in brief. (CO1)
2. (a) Describe Plasma arc welding process and its benefits. (CO2)
- (b) Describe Electro-slag welding process and its benefits. (CO2)
- (c) Describe Laser Beam welding process and its limitation. (CO2)

P. T. O.

(2)

3. (a) Describe the types of distortion in welding. What are the steps taken to control the same ? (CO3)
- (b) What is carbon equivalent ? Indicate its application. (CO3)
- (c) Explain hot and cold cracking. How can they be controlled ? (CO3)
4. (a) Explain the following : (CO4)
- (i) Cathode spot
 - (ii) Anode spot
 - (iii) Arc column
 - (iv) Ionization potential
- (b) Indicate the different modes of metal transfer in arc welding. (CO4)
- (c) What are different welding defects ? Explain their cause of occurrence. (CO4)
5. (a) Explain in brief the physics of welding arc. (CO5, CO6)
- (b) Explain Ultrasonic and Magnetic particle tests conducted for welding. (CO5, CO6)
- (c) How X-Ray testing is done of welded joints ? Explain with sketch. (CO5, CO6)